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A New Approach to the Study of Ticket Splitting

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A new solution to the ecological inference problem is used to examine split-ticket voting patterns across states and congressional districts in the 1988 elections. Earlier studies of ticket splitting used either aggregate data, which suffer from the "ecological fallacy" and threaten individual-level inferences, or survey data from small, unrepresentative samples. We produce more accurate estimates of the proportions of voters splitting their ballots in each state and district, which enables us to examine variations across geographical units. We also clarify the connection between ticket splitting and divided government and test several competing theories about the causes of both. We find, contrary to balancing arguments, that voters are not intentionally splitting tickets to produce divided government and moderate policies. In most cases split outcomes are a by-product of lopsided congressional campaigns that feature well-funded, high-quality candidates versus unknown competitors.

The 1990s have witnessed a flurry of research on split-ticket voting and divided government in the United States (Alesina and Rosenthal 1995; Alvarez and Schousen 1993; Beck et al. 1992; Born 1994; Cox and Kernell 1991; Fiorina 1996; Frymer, Kim, and Bimes 1997; Galderisi 1996; Jacobson 1990). Both are more common features of the political landscape today than in previous periods of American history and thus warrant this scholarly attention. Fifteen of the last 22 national elections have produced divided government, and surveys show that split-ticket voting rates are higher than in previous decades. There is little consensus on what causes ticket splitting. Some researchers argue that it is done intentionally in an attempt to produce the proper mix of policies or keep the political parties in check (Alesina and Rosenthal 1995; Fiorina 1996; Ingberman and Villani 1993). Others believe it is the unintentional by-product of other factors, such as the competitiveness of congressional elections (Jacobson 1997; Kimball 1997), the blurring of ideological boundaries between the parties (Born 1994; Kimball 1997; Soss and Canon 1995), and ballot mechanisms (Beck 1997; Campbell and Miller 1957). Whatever the eventual answer, it is clear that scholars need to rethink voting behavior to produce a theory that explains ticket splitting (Sundquist 1988). Indeed, Fiorina (1996, 129) claims in his definitive *Divided Government* that "ticket-splitting and other aspects of the relationships between

voting for different offices should become a central focus of research."

We take up Fiorina's charge by using a new statistical method (King 1997) to produce more accurate estimates of the direction and magnitude of split-ticket voting in each state and congressional district in 1988. We apply the method to presidential and congressional voting behavior by estimating president/House and president/Senate ticket splitting. The estimates are then used in a series of analyses to compare our ecological estimates to other estimates and to explain why ticket splitting is so common in some places and nonexistent in others. The 1988 elections are an ideal context in which to study split-ticket voting. Unlike earlier periods in American electoral history, which produced divided government during midterm elections, the 1980s witnessed split outcomes in presidential years as well. In fact, it was the simultaneous election of George Bush with the maintenance of the Democratic status quo in Congress in 1988 that interested many researchers in split-ticket voting (Fiorina 1996; Jacobson 1990).

Though this case is theoretically appropriate, inferences about the direction and magnitude of ticket splitting have been hindered because of data limitations. Aggregate measures have been used in the literature for several decades but are flawed by the well-known "ecological fallacy," which refers to the hazards of making individual-level inferences from aggregate data. Survey data also have been used, but national samples are not large enough to make inferences about voting behavior in particular districts. We employ King's technique to produce more accurate estimates of split-ticket voting within districts and states. Unlike earlier studies based on just a few data points (indexed by time), our data set contains hundreds of observations (indexed by district), making it easier to generalize. We demonstrate the validity of these estimates and then combine them with other data to determine what causes American voters to split tickets. We find strong evidence for the contention that candidates—not voters—are primarily responsible for producing divided government. Most variation in the

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extent of ticket splitting across districts and states is not due to policy balancing on the part of voters.

It is worth studying ticket splitting at the district and state level because election outcomes are determined within these electoral units (Wright 1989). Voters within these geographical boundaries, not the national population, determine the composition of Congress and the Electoral College. Our approach allows us to link ticket splitting directly with divided government because we can compare levels of ticket splitting to the election outcome in each state or district.

USING ECOLOGICAL INFERENCE TO STUDY TICKET SPLITTING

Given the ubiquity of divided government in American national politics, it is important to understand how many voters split their ballots and why. Accurate estimates will help us understand the extent to which divided government is the will of the voters. Why voters split tickets is a continuing research question in studies of electoral behavior. Unfortunately, the data used to answer the question have not been up to the task. Here, we consider the two traditional approaches—using individual survey and aggregate election data—and demonstrate why the new method is an improvement.

In addition to King's method, several other techniques developed in political science, sociology, statistics, and other fields may be helpful in drawing individual inferences from aggregate data. Perhaps the earliest is the "method of bounds" (Duncan and Davis 1953), which uses a deterministic relationship between aggregate quantities to narrow the range of possible values of the individual-level parameters of interest. Some have applied that method while adding external assumptions or substantive knowledge to narrow even more the logical bounds around the individual-level parameters (Achen and Shively 1995; Shively 1991; Sigelman 1991). The most commonly used ecological inference method is probably Goodman's (1959) regression technique. Though Goodman warned against its use, one may estimate individual-level parameters though simple linear regression, assuming that the parameters do not vary across districts (or vary randomly across districts around their global values).

Approaches to the ecological inference problem have become remarkably complex since that time. Achen and Shively (1995) survey the latest techniques, many of which allow for some relaxation of Goodman's constancy assumption or incorporate other substantive assumptions about the individual-level parameters and their relationships with aggregate-level variables. Most of these techniques estimate only one global set of individual-level parameters, however, not a pair for each electoral district, which is essential for our purposes. King's method provides district-level estimates of the quantities of interest and the uncertainty associated with them without making unreasonable assumptions about the constancy of the parameters we are estimating. It also permits us to work *between* the individual and aggregate levels of analysis, which helps

us link ticket splitting to split-party election outcomes. In addition, while others have noted a connection between the method of bounds and Goodman's regression (Achen and Shively 1995), King's approach may be the first to combine the insights of both earlier methods.

Individual Data

As with other voting studies, analyses of ticket splitting rely primarily on survey data. The National Election Study (NES), for instance, routinely asks a sample of Americans how they voted in the most recent elections. These data have been informative for examining the national trend in ticket splitting over the last 45 years and some of its determinants. A serious problem is that one cannot decompose national survey samples into smaller units of aggregation, such as states or voting districts, because of the number of cases required for such an analysis (Voss, Gelman, and King 1995). The NES interviews about 2,000 people each election year; so relatively few respondents come from each state, and even fewer are drawn from any given House district. This makes estimates inefficient, if they are possible at all. Furthermore, a representative national sample usually is not representative of voters from each state or congressional district (Westlye 1991).¹

Compounding the problem is the potential for inaccurate reports of voting behavior in surveys (Abramson and Claggett 1986; Campbell et al. 1960; Silver, Anderson, and Abramson 1986; Wright 1993). Proportions based on survey data are only unbiased if respondents report their voting behavior accurately.² For example, researchers have pointed out that the NES overstates support for the winning (Wright 1990) and incumbent (Eubank and Gow 1983; Gow and Eubank 1984) congressional candidates.³ This is due either to forgetfulness over time, social desirability pressures, or a question wording artifact (Box-Steffensmeier and Jacobson 1995; Jacobson and Rivers 1993; Wright 1990). Others argue that surveys tend to overstate voter turnout (Abramson and Claggett 1986; Campbell et al. 1960; Silver, Anderson, and Abramson 1986). Thus, a study of voting data at an aggregated level is a worthwhile supplement to self-reported voting patterns derived from surveys.

¹ Some may advocate the use of exit polls to estimate ticket splitting within each state and district, but these usually do not include data to parcel respondents into congressional districts, and until recently many exit polls did not ask people to report their House vote. In any event, the representativeness of respondents in exit polls is frequently in question (Asher 1995).

² Technically, estimates may be unbiased if reporting errors are random. Systematic misreporting in a particular direction induces bias.

³ To the extent that incumbency contributes to ticket splitting (Alvarez and Schousen 1993; Born 1994), a proincumbency bias means that the NES may overestimate the extent of split-ticket voting in national elections in the 1980s (Wright 1990).

TABLE 1. Voting in Maryland's Sixth Congressional District, 1988				
Presidential Voting Decision	House Voting Decision			Total
	Byron (D)	Halsey (R)	Abstain	
Dukakis (D)		<i>Dukakis splitters</i>		82,781
Bush (R)	<i>Bush splitters</i>			158,808
Total	166,753	54,528	20,308	241,589

Aggregate Data

Aggregate election data also have been used to examine split-ticket voting. Burnham's (1965, 1970) comprehensive analysis of American political realignments cites ticket splitting as a sign of large-scale political change. His operationalization of split-ticket voting is the simple difference between the highest and lowest Democratic percentage of the two-party vote among races within each state. Burnham admits that his indicator is "measured rather crudely," although it was one of the best available at the time. As is well known, the problem with such a measure is that making individual inferences from aggregate data may lead to false conclusions. Furthermore, Burnham's method can only estimate total ticket splitting, not its direction; the method cannot distinguish those who select a Republican president and a Democratic congressional candidate from those who choose a Democratic president and a Republican for Congress.

Rusk (1970) uses the same aggregate measure while pointing out two of its flaws. First, "it would not detect mutual crossovers between the two parties" (p. 1224); second, differences in the type of election being held in the states the same year threaten interstate comparisons, a problem Rusk calls the "race composition factor." The latter is a concern, but consider the more immediate ramifications of mutual crossover. Even if two states have the same contests on the ballot, Burnham's measure understates ticket splitting, since ballots split in opposite directions cancel out one another. For instance, a value of zero may mean that no split-ticket voting occurred (as Burnham and Rusk assume) or that every voter is casting a split ticket. The truth is likely to lie somewhere in between. Much may be happening at the individual level, but aggregate data cannot reveal it, a consequence of the ecological fallacy that has plagued social science for decades (King 1997; Robinson 1950).

Another aggregate technique relies on the two-office split result, which assumes that a difference in party outcomes for two offices in the same district is evidence of widespread ticket splitting. The increasing proportion of congressional districts with split results in presidential and House contests is used to infer that ticket splitting has increased over time (Cummings 1966; Fiorina 1996). But Feigert (1979) shows that this technique can produce inaccurate estimates and instead finds a high correlation between split districts and ballot roll-off.⁴ Below, we show that once the direction

⁴ Feigert (1979) actually recommended the use of ecological inference techniques in further study of ticket splitting but lamented that

of ticket splitting is properly estimated, there is a conditional relationship between split-ticket voting and split outcomes.

With the development of a solution to the ecological inference problem, we can produce district- and state-level estimates of the proportion of voters casting split ballots that avoid the problems described above. Since we use aggregate data, survey misreporting is not a concern; only collective (and anonymous) votes are needed. In addition, King's method allows us to account for both types of ticket splitting, thus avoiding the crossover problem in previous aggregate measures. Before turning to the estimation itself, we briefly describe the ecological inference problem.

The Problem as a 2 × 3 Contingency Table

The general ecological inference problem may be conceptualized as a standard contingency table with missing data. The marginals of the table are known, since they are based on aggregate data, but the cell percentages are unknown (King 1997; Markus 1974). Table 1 lays out the ecological inference problem in estimating ticket splitting, using 1988 congressional election results from Maryland's Sixth District as an example. We choose this district for illustration because it produced a split-party result that matched the national pattern: a Republican (George Bush) at the presidential level and a Democrat (Beverly Byron) at the congressional level. Since the presidential contest usually appears at the top of the ballot and is the more visible race, we assume that voters make their presidential choice first. Voters then have three options in the congressional races that appear farther down the ballot: (1) vote Democratic, (2) vote Republican, or (3) abstain.⁵ From election returns we know how many voters in Maryland's Sixth District chose Bush in the presidential contest (158,808), and we know how many voters selected Byron, the Democratic candidate for the House (166,753), but we do not know how many of the Bush voters split their ballot by also selecting Byron. Presumably, this was common, since each candidate carried the district by a hefty margin. We do know that the number of Bush voters who split their ballot ("Bush splitters") is between 83,972 (the Byron

valid methods did not exist at the time. While ecological inference techniques have been used to examine the extent to which voters switch parties from one election to the next (Achen and Shively 1995; Gosnell 1937), to our knowledge no one has used these techniques to study ticket splitting.

⁵ We ignore votes for third-party candidates. Few voters chose such candidates in 1988.

total minus the Dukakis total, or .528 of all Bush voters in the district) and 158,808 (all Bush voters), a wide interval. Similarly, we have reason to believe that slightly more than 20,000 presidential voters "rolled off" by not casting a vote in the House race, but we do not know whether abstention in the House contest was more prevalent among Bush or Dukakis voters.

Our estimation of ticket splitting is a two-stage process that divides Table 1 into two 2×2 tables that are analyzed sequentially for every district. First, we estimate the fraction of Dukakis and Bush voters who also cast a vote in the House contest (or the proportions that do not fall in the "Abstain" column). Those estimates are then combined with the other known aggregate information to estimate ticket splitting ("Bush splitters" and "Dukakis splitters").⁶ It is important to control first for ballot roll-off in the congressional contest, because voters who abstain from that race cannot split their ballot. In addition, if there are partisan differences in roll-off, then aggregate measures of ticket splitting that fail to account for roll-off are biased.⁷

When discussing our estimates of split-ticket voting, we will use a two-letter abbreviation to indicate the four possible voting patterns; the first letter is the party of the presidential candidate chosen, and the second letter is the party of the congressional candidate chosen. For instance, an RD voter is one who voted for Bush and the Democratic congressional candidate.

DATA AND METHOD

The main data set consists of only the number of votes received by the presidential, House, and Senate candidates in each congressional district or state (from Barone and Ujifusa 1989) to provide the known quantities in Table 1.⁸ Voters in 1988 could have split their ballot in one of two ways: Bush and a congressional Democrat, or Dukakis and a congressional Republican. "Bush splitters" denotes the former, and "Dukakis splitters" denotes the latter. Ultimately, we are more

interested in Bush splitters because of the divided government pattern of the 1980s.⁹

The first run included only presidential and House data, with 74 uncontested House races excluded from the procedure. In the uncontested districts, voters only had one choice for the House; the exclusions do not affect the estimates for the other districts. In addition, a South dummy variable was used as a covariate to account for possible aggregation bias and to improve the estimates. Because the degree of split-level partisanship in the South is unique, we suspect that aggregating votes may conceal more ticket splitting in the South than elsewhere. The second run produced estimates of president/Senate ticket splitting within states. Since there were only 33 Senate races in 1988, estimates based on so few observations would be inefficient. Thus, the House data were included to improve the Senate estimates.

Our estimates reveal that ticket splitting varies substantially across districts and states, even within the same election year. After removing the uncontested House races, the proportion of Bush splitters in a district ranges from .03 to .90, with a standard deviation of .19. The proportion of Bush splitters was under .1 in 23% of the districts, but it exceeded .5 in 27.3% of the districts. There was less variation in the district-level fraction of Dukakis splitters, which ranges from .02 to .72, with a standard deviation of .14. The proportion of Dukakis splitters was under .1 in 42.1% of the districts and above .3 in 29.6% of the districts. The relationship between the two types of ticket splitting is negative and strong ($r = .94$); in districts with many Bush splitters, there were few Dukakis splitters. Nevertheless, some form of ticket splitting was quite common in many congressional districts in 1988.

Because these are maximum likelihood estimates, King's method also produces standard errors for the two ticket-splitting parameters for each congressional district. In this case it yields fairly precise estimates of ticket splitting. The mean standard error of our district-level estimates for Bush splitters is .04, while for Dukakis splitters it is .05. Thus, traditional 95% confidence intervals drawn around our estimates would cover a range of about 15–20 percentage points.

THE VALIDITY OF OUR ESTIMATES

The Relationship between Split-Ticket Voting and Divided Outcomes

Does ticket splitting matter if only a minority of voters are involved? Can a small proportion of voters who split their ballot produce divided government? Are higher levels of ticket splitting associated with split outcomes when votes are aggregated at the district level? Table 2 provides the mean level of each type of

⁶ With King's *Ezi* program (version 1.21), we used the EI command to make the first-stage estimates and the EI2 command to make the second-stage estimates. See King (1997, chapter 15) and the Appendix for a more detailed description of this method.

⁷ According to our estimates, Democrats were somewhat more likely than Republicans to abstain from the House contest. In addition, ballot roll-off was quite high in some congressional districts, especially if an incumbent ran unopposed or faced token opposition. As it turns out, our ticket-splitting estimates did not change substantially when we did not control for ballot roll-off. For an example that controls for roll-off in studying racially polarized voting, see Palmquist and Voss (1996).

⁸ Eleven districts (from Arkansas, Florida, Louisiana, and Oklahoma) that featured uncontested House races were excluded from our analysis because election totals were not available. The state either did not put the House contest on the ballot (as in Louisiana when a candidate won a majority of the vote in the open primary) or did not tabulate the House vote in these districts. Survey respondents from these districts (61 total) were dropped from the 1988 NES sample for the comparisons of ticket-splitting estimates presented here. The Appendix offers a brief description of King's ecological inference method.

⁹ The RD pattern of national government common in the 1970s and 1980s is remarkable. Indeed, according to Fiorina (1989, 140), "the emergence of a two-tier system in which Republican presidential majorities coexist with Democratic congressional majorities is the single most important feature of contemporary politics."

TABLE 2. Mean President/House Ticket Splitting by District Outcome, 1988				
Ticket-Splitting Type	District Outcome			
	RR	RD	DR	DD
Bush splitters	.11 (.08)	.57 (.24)	.09 (.05)	.59 (.24)
Dukakis splitters	.41 (.22)	.09 (.06)	.43 (.13)	.07 (.05)
Number of districts	163	135	12	125

Note: Standard deviations of the point estimates are in parentheses.

President/House ticket splitting in different types of districts in 1988. Ticket splitting by Bush voters is quite high (.57) in districts that chose Bush and a Democrat but low (.11) in RR districts. Nevertheless, ticket splitting and divided outcomes are not entirely synonymous, though they are sometimes treated as such. Substantial numbers of voters split their ticket even in districts where candidates of the same party are elected. For example, .59 of Bush voters (on average) split their ticket in DD districts. Notice that Dukakis splitters are most common in DR and RR districts, and Bush splitters are most frequent in RD and DD districts. In accord with the argument we make below, this suggests that the primary determinant of ticket splitting is the local congressional race, not a national mandate for split control.

Consider that Bush can be expected to defeat Dukakis in more conservative districts. When a Democratic congressional candidate wins in such a district, it is likely due to a high rate of ticket splitting among Bush voters. If divided outcomes and split-ticket voting were unrelated, then one would expect no connection between the fraction of Bush splitters in a district and an RD election result.

To test this connection, we examine the relationship between the level of ticket splitting among Bush voters and the probability of districts having a unified Republican electoral outcome (RR) or a divided outcome (RD). The estimation is a logit model in which the dependent variable is the probability of selecting a Democratic congressional candidate, conditional on the selection of a Republican for president. The results, found in Table 3, show that the relationship between ticket splitting among Bush voters and the election of a Democrat to the House is remarkably

TABLE 3. Logit Equation Predicting Divided Outcomes (RD versus RR Districts) by Split-Ticket Voting	
Variable	b
Bush splitters	63.84*** (16.29)
Constant	-18.71*** (4.77)
n	298
Correctly predicted	98.3%
Log-likelihood	-11.82

Note: Standard errors are in parentheses. ***p < .001, two-tailed t test.

strong. The classification rate of the one-variable model improves upon the naive baseline significantly, mispredicting less than 2% of districts based only on our knowledge of ticket splitting among Bush voters. The logit equation predicts that if more than .29 of the Bush voters split their ballot, then there is a better than 50% chance that the district will elect a Democrat to the House. Thus, both the direction and frequency of ticket splitting determine divided outcomes. This strong association between divided outcomes and our estimates of ticket splitting further validates the use of King's method in this case.

Comparisons with Individual Estimates

It is useful to compare our ecological-based estimates with those from a more traditional individual-level source. Table 4 presents both for a quick comparison, along with estimates using Goodman's regression to show how much we improve on previous ecological inference methods. Using the 1988 NES, the estimated proportion of Bush voters who also chose a Democrat for the House (Bush splitters) is .341, as compared to our estimate of .331. The corresponding proportion of Dukakis splitters is a little less than half that, at .151, as compared to our ecological estimate of .198. At the national level our ecological estimates of ticket splitting are close to those produced by the NES, with smaller standard errors to boot. The NES estimates have standard errors of about .025, or 2.5%, so the 95% confidence interval around the national point estimates includes our estimates, making them statistically indistinguishable.

For the 33 Senate races in 1988, split-ticket voting was about as common as in the House contests. Roughly one in three Bush voters chose a Democratic Senate candidate, and about one in five Dukakis voters chose a Republican for the Senate. To compare the ecological estimates with individual data, we estimated split-ticket voting from the 1988 Senate Election Study (SES), which drew representative samples of about 75 people from each state with a Senate race that year. As Table 5 indicates, the ecological estimates of president/Senate ticket splitting are a bit lower than those produced by the SES, but we did not weight the SES data by state population, which may explain why those estimates are higher. Also, small states (where ticket splitting was more common in 1988) are overrepresented in the SES. Finally, while the Goodman estimates are close to the other two in this case, the large standard errors indicate a high level of uncertainty in the Goodman estimates.

EXPLAINING VARIATION IN TICKET SPLITTING ACROSS DISTRICTS AND STATES

In the previous section we found tremendous variation in ticket splitting across congressional districts and states. In this section we attempt to explain why split-ticket voting is more common in some electoral units than in others. We present results of multivariate

TABLE 4. Three Estimates of President/House Ticket Splitting, 1988			
Source	Proportion Splitting Ballots		
	Bush Voters	Dukakis Voters	All Voters
Ecological estimate	.331 (.006)	.198 (.007)	.271
NES estimate	.341 (.021)	.151 (.016)	.254
Goodman estimate	.033 (.032)	-.108 (.037)	.018

Note: Standard errors are in parentheses.

analyses using our district and state point estimates of split-ticket voting as dependent variables. Our analysis provides a direct comparison of competing theories of split-ballot voting.

On the one hand, intentional theories are based on the notion that some voters prefer divided government and pursue it. Ideological moderates purposely vote for presidential and congressional candidates of different parties so that the separate branches of government will check each other (Alesina and Rosenthal 1995; Erikson 1988; Fiorina 1996; Jacobson 1990). Thus, split-ticket voting is consciously used to produce divided government. Some survey respondents claim that they prefer divided to unified control of government (Fiorina 1996). Ticket splitting may be a viable strategy for balancing the extreme positions of opposite party candidates, thereby producing moderate policies favorable to the median voter.

On the other hand, unintentional theories posit that split-ticket voting is a by-product of other forces, such as weak partisan attachments (Petrocik and Doherty 1996), ballot mechanisms (Beck 1997; Campbell et al. 1960; Campbell and Miller 1957), and the level of competition in congressional contests (Jacobson 1990). The lack of intention is attributed only to voters; candidates play an important role in producing split-ticket voting. For example, Bush voters in a district with an unchallenged Democratic candidate for Congress in 1988 have little choice but to vote for the Democrat or abstain. Furthermore, as party attachments in the electorate have weakened, campaigns have become more candidate centered (Wattenberg 1991), making it more likely that voters will choose candidates from different parties, even if not motivated by the specific goal of divided government. Jacobson (e.g., 1997) demonstrates that well-funded candidates with previous political experience are most adept at attracting voters from the opposite party. Voters simply may select the most visible or most appealing candidate for each office; given the low level of competition in many congressional contests, this decision rule often leads to ticket splitting.

Since the direction of ticket splitting is important in determining the election outcome, we treat the two types of ballot splitting (RD versus DR) separately. We estimated two linear regression equations to determine the factors that influence the fraction of Bush voters who select a Democrat for Congress and the fraction of Dukakis voters who select a Republican for Congress, using states or districts as our units of analysis. We focused most of our attention on RD splitting, since the 1988 election produced a Republican president and a Democratic Congress. We included two explanatory variables to account for the nature of the congressional campaign. One is a dummy variable for the party of the incumbent, who often develops a personal bond with constituents that crosses party lines (Fenno 1978). We expect that RD ticket splitting will be higher in contests with a Democratic incumbent and that DR splitting will be more common when the incumbent is Republican. The other variable is a measure of the Democratic candidate's spending as a fraction of total spending in the congressional contest. Campaigns are contests of competing messages crafted to attract voters, and the spending variable measures the extent to which the Democrat's campaign message is dominating that of the Republican. In addition, the spending measure allows us to control for the fact that campaign costs vary from one district to another. We expect that campaign spending will be positively associated with RD ticket splitting and negatively associated with DR splitting.

We also included a dummy variable for districts or states in the South. Since World War II, the South has supported Republican presidential candidates and Democratic congressional candidates more often than the rest of the country. Frymer (1994) and Frymer, Kim, and Bimes (1997) argue that RD ticket splitting has been more common in the South because Democratic candidates for Congress have been able to position themselves toward the conservative end of the ideological spectrum, thus capturing the support of many Republican presidential voters. It is argued that

TABLE 5. Three Estimates of President/Senate Ticket Splitting, 1988			
Source	Proportion Splitting Ballots		
	Bush Voters	Dukakis Voters	All Voters
Ecological estimate	.285 (.017)	.187 (.020)	.237
SES estimate	.333 (.020)	.255 (.022)	.298
Goodman estimate	.359 (.190)	.262 (.226)	.313

Note: Standard errors are in parentheses.

TABLE 6. Regressions for RD Ticket Splitting						
Variable	President/House			President/Senate		
	OLS <i>b</i>	WLS <i>b</i>	OLS β	OLS <i>b</i>	WLS <i>b</i>	OLS β
Ideological distance	—	—	—	-.029 (.039)	-.042 (.049)	-.088
Democratic incumbent	.107*** (.015)	.095*** (.023)	.279	.096* (.044)	.077 (.055)	.297
Spending ratio	.350*** (.021)	.443*** (.031)	.664	.290** (.084)	.379** (.105)	.511
Ballot format	-.032*** (.008)	-.074*** (.011)	-.083	-.067 (.037)	-.071 (.045)	-.197
South	.052*** (.009)	.116*** (.012)	.116	.037 (.046)	.017 (.053)	.090
Constant	.065*** (.008)	.049*** (.011)		.138 (.082)	.132 (.098)	
Standard error	.073	.101		.093	.108	
<i>n</i>	361	361		32	32	
Adjusted <i>R</i> ²	.855	.845		.678	.686	

Note: Uncontested races are excluded. Cell entries are regression coefficients with standard errors in parentheses. **p* < .05, ***p* < .01, ****p* < .001, two-tailed *t* test.

moderate to conservative voters use simple proximity decisions to select congressional Democrats and presidential Republicans. We likewise expect RD ticket splitting to be more common in the South.

It has been suggested that ticket splitting is less common in states with a straight-party option on the ballot (Beck 1997; Campbell and Miller 1957; Rusk 1970). As Campbell et al. (1960, 275) note, “any attempt to explain why the voter marks a straight or split ballot must take account of the physical characteristics of the election ballot.” They found that split-ticket voting was about 8% higher in states without a straight-party option in the 1950s. In 1988, 20 states had a ballot that allowed voters to pull one lever or check one box to vote a straight ticket (Brace 1993). Thus, we included a dummy variable for states with a straight-party option, and we expect it to be negatively associated with both types of ticket splitting.

The intentional theories offer few hard predictions, but Fiorina (1996) presents a clear hypothesis. Based on a simple spatial voting model in which moderate voters are located between an extremely conservative Republican Party and an extremely liberal Democratic Party, he finds that classic proximity voting may no longer be rational when candidates are far from a voter’s ideal point. Instead, candidates from opposite parties are chosen for different offices so that they will be required to compromise and produce moderate policies after taking office. A key tenet of this argument, which Fiorina says is “probably the most important point,” is the following prediction: “When the parties are relatively close, near the center of gravity of the electorate, ticket-splitting declines. When the parties move away from each other . . . they open up a large policy range in which ticket-splitting is the voter response” (Fiorina 1996, 81).¹⁰

We have a unique opportunity to test this hypothesis

with available Senate data. The SES asked voters in states with Senate races to place the presidential and senatorial candidates on a seven-point ideological scale.¹¹ Taking the mean placements of the four candidates in each state, we computed the mean distance between Bush and the Democratic Senate candidate. Based on the prediction from Fiorina’s balancing perspective, we should find a positive relationship between this distance and RD voting. In other words, RD ticket splitting should increase as the distance between Bush and the Democratic Senate candidate increases.

Table 6 presents regression results using RD ticket splitting as the dependent variable. The first column reports traditional ordinary least squares (OLS) estimates and the second column presents weighted least squares (WLS) estimates to account for the differing standard errors associated with our point estimates of ticket splitting.¹² In the WLS analysis, observations are weighted by the inverse of these standard errors, thus giving greater weight to observations with more precise estimates of ticket splitting. Because OLS and WLS produced similar results, we retained the OLS estimates because of their intuitive interpretations. The final column reports standardized OLS coefficients to assess the relative effects of the explanatory variables. Just a few independent variables do an impressive job of explaining variation in ticket splitting. Though parsimonious, the OLS model explains 86% of the variation in the President/House results and 68% of the variance in the President/Senate results.¹³ This amount

¹⁰ Fiorina models voter behavior as if it were conscious. Our test appropriately examines voter behavior without peering inside the “black box” to check the assumptions about process.

¹¹ “Unless we have data on both the Democratic incumbent and her Republican challenger, and unless we can put the presidential candidates and the voters on the same scale, we cannot differentiate among such different situations and hence decide whether ticket-splitting is consistent with an ideological consistency or ideological balancing model” (Fiorina 1996, 155). The SES data satisfy these criteria.

¹² The substantive results are unchanged if two-sided tobit models are estimated to account for a dependent variable censored at zero and 1.0.

¹³ The Senate analysis is based on 32 states because the 1988 SES did

TABLE 7. Regressions for DR Ticket Splitting						
Variable	President/House			President/Senate		
	OLS <i>b</i>	WLS <i>b</i>	OLS β	OLS <i>b</i>	WLS <i>b</i>	OLS β
Ideological distance	—	—	—	.002 (.036)	−.0004 (.044)	−.007
Republican incumbent	.085*** (.011)	.101*** (.014)	.300	.044 (.036)	.008 (.043)	.200
Spending ratio	−.252*** (.015)	−.265*** (.018)	−.659	−.225*** (.059)	−.315*** (.068)	−.610
Ballot format	.003 (.006)	.005 (.006)	.011	.034 (.028)	.032 (.033)	.153
South	.007 (.006)	.005 (.008)	.023	−.001 (.032)	−.004 (.039)	−.005
Constant	.309*** (.013)	.313*** (.016)	—	.277*** (.070)	.353*** (.084)	—
Standard error	.052	.059		.069	.078	
<i>n</i>	361	361		32	32	
Adjusted <i>R</i> ²	.863	.852		.585	.602	

Note: Uncontested races are excluded. Cell entries are regression coefficients with standard errors in parentheses. ****p* < .001, two-tailed *t* test.

of explained variance is high in comparison to other voting behavior models but is especially surprising given that the dependent variable is merely a point estimate based on ecological data.¹⁴

The House and Senate equations differ only in that the Senate includes a measure of ideological distance between Bush and the Democratic senatorial candidate. Note that its coefficient is negative and statistically insignificant. The lack of significance is likely due to the small number of cases (*n* = 32) and corresponding degrees of freedom since the zero-order correlation between RD ticket splitting and the ideological difference between the candidates is −.370 (*p* < .05). Contrary to the balancing hypothesis, this means that RD ticket splitting does not increase as the parties polarize and that it may actually decrease. Split ballots are most common when Bush and the Democratic candidate are nearest each other, blurring their ideological differences enough to make partisan considerations in voting behavior less important (Frymer 1994; Kimball 1997). This finding fits with the Downs (1957) expectation that, in an attempt to win votes, party platforms will be ambiguous and overlapping.

In the House equation, RD voting is more common in the South. According to the OLS results, about 5% more Bush voters chose the Democratic House candidate in the South than elsewhere. Assuming for the moment that Democratic candidates are relatively conservative in the South, the distance between their position and that of Bush should be relatively small (Frymer, Kim, and Bimes 1997). Despite this, ticket splitting is higher, not lower. Again, intentional theo-

ries that attribute ideological balancing motives to split-ticket behavior are not supported.

The other variables in the equation are all significant and have the expected signs. The presence of a Democratic incumbent boosts RD ticket splitting by roughly ten percentage points in both House and Senate contests, holding other variables constant. Furthermore, a straight-party ballot mechanism decreases RD splitting about three percentage points in the House races and almost seven percentage points in the Senate contests. The greatest influence by far, however, is campaign spending. Recall that this variable is operationalized as the Democratic proportion of all congressional campaign spending. The OLS coefficient of .35 indicates that a Democratic House candidate who matches the Republican opponent dollar for dollar can increase RD ticket splitting by a whopping 17.5 percentage points over a Democrat who spends no money on the campaign. The parameter estimate of .29 in the Senate equation indicates that a Democratic senatorial candidate who matches the Republican opponent in spending will increase RD voting by nearly 15 percentage points over a Democrat who spends no money on the campaign. Furthermore, the relative effect of spending is overwhelming, as the standardized regression coefficient is roughly twice as large as that of the next most important variable.¹⁵

not obtain candidate placement information from Nevada respondents.

¹⁴ One may argue that our ideological distance measure will not work properly if both Senate candidates fell on the same side of the median voter in a particular state. Using the SES, however, in almost every state the mean ideological placement for respondents fell in between the mean placements of the two Senate candidates (Erikson 1990).

¹⁵ One may object that we have not accounted for the effect of the presidential candidates on ticket splitting. It is possible that some voters defect from their identified party in the presidential contest. To control for the presidential contribution, we included Bush's fraction of the vote in the regression models presented here. This did not improve the fit of the models and did not alter the effects of the other variables. In addition, it produced substantively weak and inconsistently signed regression coefficients. For example, the Bush variable had a positive coefficient (*p* = .03) in the first House model in Table 6, but a negative coefficient (*p* = .24) in the first House model in Table 8. The effect in Table 6 (although substantively small) suggests that Bush contributed to ticket splitting by attracting some Democratic votes.

Before exploring this finding, let us consider a similar regression, shown in Table 7, in which the dependent variable is the fraction of Dukakis voters who chose a Republican congressional candidate. The independent variables remain the same except that ideological distance is now measured between Dukakis and the Republican senatorial candidate, and the incumbency dummy now denotes a Republican. The analysis again points to the importance of the candidates in the congressional contest. For example, a Republican incumbent increases DR voting in House contests by about nine percentage points, other things being equal. In addition, spending remains the most critical factor, whether judged by the standardized coefficient or substantively, by interpreting the unstandardized coefficient. As a Republican candidate's share of total congressional campaign spending goes from zero to 100%, DR ticket splitting increases by about 25 percentage points in the House contest and about 23 percentage points in the Senate race. That is, the percentage of Dukakis splitters may jump by more than 20 points based on congressional spending patterns alone. Certainly, campaign spending is a proxy for other factors, such as the relative quality and experience of the candidates, but it also is critical to ticket splitting. Candidates spend most of their funds on advertising to increase their visibility and emphasize positive, familiar images in the minds of voters (Hernson 1997; Jacobson 1997). To the extent that a congressional candidate can dominate advertising and frame the campaign, s/he will attract more votes from the opposite party.

With regard to the Senate, as in Table 6, the results for ideological distance in Table 7 reveal that this is not a statistically significant influence on DR ticket splitting. In our model, increased distance between Dukakis and the Republican senatorial candidate does not produce more ticket splitting, contrary to predictions of the balancing perspective. Furthermore, DR voting is no more common in the South than in other regions of the country. The South in the 1980s was unique for one type of ticket splitting: a Republican for president and a Democrat for Congress. Finally, while a straight-party ballot option reduced RD ticket splitting in both Senate and House races, it did not have a noticeable effect on DR outcomes in 1988.

A more interesting setting may be the House elections. After all, government was divided along president/House lines throughout the 1980s, while both Senate and White House were controlled by Republicans for six years. It is more difficult for voters to split government control using their Senate vote, because there are only 33 races in a given election year. Our test of ideological balancing in presidential and senatorial elections did not support intentional theories, but comparable survey data were not available for the House. Instead, we used ideological scores for Democratic incumbents, since Democrats were a large majority in 1988, although this necessarily limits our test to RD splitting.¹⁶ Our measure is the Poole and

¹⁶ RD splitting was the more common form in 1988 and corre-

TABLE 8. Factors Influencing RD Splitting in Districts with a Democratic Incumbent

Variable	OLS <i>b</i>	WLS <i>b</i>	OLS β
Conservatism of Democratic Incumbent	.132** (.039)	.255*** (.053)	.22
Spending ratio	.327*** (.041)	.508*** (.056)	.47
Experienced Republican challenger	-.014 (.013)	-.031 (.018)	-.07
Ballot format	-.065*** (.012)	-.122*** (.016)	-.31
South	.041** (.016)	.092*** (.020)	.17
Constant	.250*** (.039)	.170** (.055)	—
Standard error	.083	.106	
<i>N</i>	190	190	
Adjusted <i>R</i> ²	.369	.655	

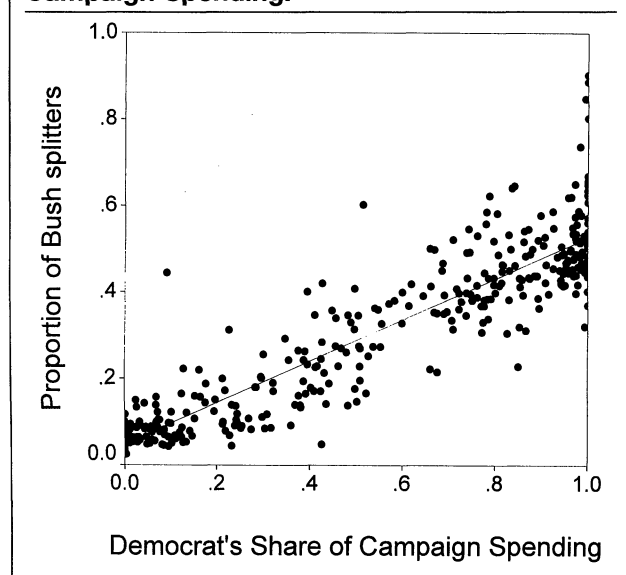
Note: Analysis only includes districts with a challenged Democrat. Cell entries are regression coefficients with standard errors in parentheses. ****p* < .01, ***p* < .001, two-tailed *t* test.

Rosenthal (1997) NOMINATE score, which is based on statistical analyses of roll-call voting.¹⁷ In contrast to the SES data, based on citizen surveys, NOMINATE is more clearly exogenous to the vote decision. We used first-dimension point estimates from the 100th House to measure the ideological position of incumbent candidates. Values range from -1 (most liberal) to +1 (most conservative). If voters are more likely to split their ballot when the parties are polarized, then the ideological position coefficient ought to be negative. In other words, if motivated to balance extreme positions, then Bush voters would be more likely to split their ticket for a liberal than a conservative Democratic House candidate.

We included this ideology measure in a regression equation similar to the ones in tables 6 and 7, except that the incumbent dummy was removed by necessity. We also added a dummy indicating the presence of an "experienced" Republican challenger in the House contest (one who previously had held elective office or had run for a House seat). Table 8 reports the results. Though spending remains the most important influence on RD voting, the extremism of the Democratic incumbent plays a part as well. The coefficient is

sponded to the election of Bush and a Democratic House. When we examined DR splitting, the results were similar to those presented in Table 8: Ticket splitting by Dukakis voters was more frequent in districts with a Republican incumbent toward the liberal end of the spectrum.

¹⁷ NOMINATE is a maximum likelihood technique that uses roll-call data ("yeas" and "neats") to estimate the ideal points of legislators. It is based on a multidimensional spatial model in which legislators take roll-call positions that are nearest their ideal point with some error, thus making it probabilistic (Burden 1997). As is common practice, we use first-dimension NOMINATE scores, which explain more than 80% of the variance in roll-call voting patterns. Furthermore, we use a particular dynamic variant, D-NOMINATE. Substantive results are no different if other forms of NOMINATE are used; they correlate with one another at well above .95. See Poole and Rosenthal (1997) for a comprehensive review.

FIGURE 1. Split-Ticket Voting by House Campaign Spending.

positive and significant, indicating that the proportion of RD voters increases in accord with Democratic conservatism. Assuming that Bush adopts a right-of-center position, RD voting is more likely if the House Democratic candidate is closer to Bush, which means that voters seek ideological consistency rather than balance. This is consistent with arguments made by Frymer (1994) and Petrocik and Doherty (1996).

Substantively, the coefficient on the ideological variable means that the RD proportion increases by 13 percentage points when the Democrat moves one unit in the conservative direction on the $[-1,1]$ NOMINATE interval. This is not a small effect and, in standardized terms, ranks only behind ballot format and the influence of money. Furthermore, the ideological variable reduces the magnitude of the coefficient for the South variable. This suggests that RD ticket splitting is higher in the South at least partly because the Democratic candidates position themselves closer to the conservative end of the spectrum.

The most robust finding from Table 8 and the other regression analyses is that congressional campaign spending has a dramatic influence on the percentage of voters who split their ballot. As the Democratic candidate's share of spending rises, so does the proportion of Bush voters who vote for that candidate. This relationship remains the strongest force in ticket splitting, even when controlling for such related and theoretically important variables as challenger quality. In fact, the challenger quality coefficient, while having the expected negative sign, is not statistically significant. That effect is clearly captured in the spending variable, as experienced challengers are most adept at cutting into an incumbent's spending advantage (Jacobson 1997). As we elaborate below, it is not the inherent quality of candidates that matters but their ability to represent a credible alternative to the incumbent.

To illustrate the critical role of spending, Figure 1

presents a simple scatterplot showing the share of Democratic spending and the proportion of Bush splitters in contested House districts. The least-squares regression line is superimposed. The relationship is clearly linear, positive, and strong considering the crudeness of the data ($r = .91$, $p < .001$). Though money has been shown to be an important variable in congressional campaigns (Jacobson 1997), its power to induce ticket splitting has been largely unexplored (cf. Kimball 1997).

We suggest that campaign spending increases voter defection from the opposite party by drowning out the opponent's message. As the spending proportion of the Democrat nears 100%, the Republican's self-promotion is scarcely noticed by voters. Lopsided spending patterns—near both ends of Figure 1—indicate one-sided messages in a campaign. As Zaller (1992) has shown, a campaign or other media message can have a substantial effect on public opinion when countervailing messages are absent. The dominant campaign's repeated advertising raises the number of favorable "considerations" associated with that candidate in the minds of potential voters. In the voting booth, they have little incentive to vote for the unknown candidate.

This examination of the causes of split-ticket voting at the district and state level finds that most of the influences are outside the voter's control, which supports the unintentional theories of divided government. This does not mean that divided control is an accident. Voters appear to be rationally choosing the candidates who seem the most competent, defined by the cues of incumbency and campaign spending, the two largest influences on split-ticket voting. Though ideological position is a factor when congressional incumbents are on the ballot, voters do not intentionally balance a conservative presidential candidate with a liberal congressional candidate to achieve moderate policy outcomes. Our evidence indicates that split-ticket voting actually decreases as candidates move farther apart. Voters choose sincerely in presidential and congressional campaigns but must resort to the best—sometimes the only—congressional candidate when intercandidate quality differences are large.

CONCLUSION

Amid the flurry of research on divided government and split-ticket voting, this article makes two contributions. First, it uses a new ecological inference technique to produce more accurate and efficient estimates of the direction and magnitude of ticket splitting in different states and congressional districts. Previous research had to depend on aggregate data—thereby inducing the ecological inference problem—or individual-level survey data that could not be used to estimate district-level parameters. These new estimates alone are valuable, for they demonstrate considerable variation in the direction and magnitude of ticket splitting from one electoral unit to the next. Though our application was to 1988, we believe that our conclusions apply to most modern presidential elections.

The second contribution lies in explaining the newly

Presidential Voting Decision	House Voting Decision				Fraction of All Voters
	Democrat	Republican	Vote (Subtotal)	Abstain	
Democrat (Dukakis)	β_i^b	$1 - \beta_i^b$	λ_i^b	$1 - \lambda_i^b$	X_i
Republican (Bush)	β_i^w	$1 - \beta_i^w$	λ_i^w	$1 - \lambda_i^w$	$1 - X_i$
Fraction of voters	T_i	$1 - T_i$	V_i	$1 - V_i$	1

revealed split-ticket voting patterns. Accounting for some intentional and unintentional forces that affect voting behavior, we find that the latter exert a tremendous influence. Though voters act reasonably considering the array of candidates offered to them, they apparently do not consciously choose divided government in the face of feasible alternatives. Elite decisions, including the decision of particular candidates to run for Congress, coupled with the financial advantages that most incumbents possess drive voters toward split ballots.

APPENDIX: KING’S METHOD

King’s ecological inference technique involves a combination of procedures. It begins by treating the problem as a 2 × C table in which the marginals are known from aggregate data but the cells are not. Notation for the known and unknown quantities can be found in Table A-1. First, the *method of bounds* is employed to restrict estimates of β_i^w and β_i^b , the cell quantities of interest, to a narrower region than the [0,1] interval (Duncan and Davis 1953). In our application, β_i^w is the proportion of Bush voters who split their tickets, and $1 - \beta_i^w$ is the proportion of Dukakis voters who split their tickets. Knowing these deterministic bounds increases the amount of information that can be used in a statistical model. King’s method posits a probabilistic distribution of the quantities of interest that assumes that β_i^w and β_i^b are distributed truncated bivariate normal (TBN), conditional on X_i .

Using the TBN distributional assumption, one can create a tomography plot (β_i^w by β_i^b), in which each line represents a district. The true values of β_i^w and β_i^b must fall somewhere on the line. The normality assumption helps detect a mode on the tomography plot where most of the lines cross and improves estimates by “borrowing strength” from other districts. Intuitively, the most likely values for β_i^w and β_i^b are nearest the mode, which is the peak of the three-dimensional normal bell shape.

After adding a couple of modest assumptions, five intermediate parameters that define the TBN distribution are estimated via a maximum likelihood function (i.e., two means, two standard deviations, and a correlation that make up the vector ψ). The parameters are then transformed and rescaled. Values of these parameters are drawn from the posterior distributions using importance sampling. The means of the distributions of β_i^w and β_i^b are used as point estimates, and standard errors (and confidence intervals) are based upon the variation in the simulated values. This is a real benefit, as earlier methods suffered from an inability to report accurate levels of uncertainty associated with point estimates (King, Keohane, and Verba 1994). This is the intuition; see section III of King (1997) for a complete description of the method.

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